

### === Harwell RSX Characterization Report ===

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that depending on the unique assembly of each resistor, we may characterize it via a polynomial or third-degree fit.

#### === Instrument Summary ===

Date: 2025-08-18

Model: Harwell RS-7

Value: 25  $\Omega$

S/N: RS70251

#### === Calibration Data ===

Calibrated Resistance: 25.00246  $\Omega$  at 23°C

Expanded Uncertainty of Measurement, k=2: 1.04  $\mu\Omega/\Omega$

Measurement Current: 1 mA

#### === Temperature Characterization Data ===

$\alpha$  ( $\Omega/^\circ\text{C}$ ): -2.59e-05

$\beta$  ( $\Omega/^\circ\text{C}^2$ ): -2.06e-06

Linear TCR (18-28°C): -1.01 PPM/ $^\circ\text{C}$

To calculate the resistor value at a temperature other than 23°C, use the following model where T is the desired temperature in  $^\circ\text{C}$ :

$$R(T) = R_{(23)} + \alpha(T - 23) + \beta(T - 23)^2$$

Note: This model may not be accurate outside the characterized temperature range.

| Temperature ( $^\circ\text{C}$ ) | Temperature ( $^\circ\text{F}$ ) | Resistance ( $\Omega$ ) | Deviation (ppm) |
|----------------------------------|----------------------------------|-------------------------|-----------------|
| 18.0                             | 64.4                             | 25.0025381              | 3.12            |
| 18.5                             | 65.3                             | 25.0025349              | 3.00            |
| 19.0                             | 66.2                             | 25.0025307              | 2.83            |
| 19.5                             | 67.1                             | 25.0025254              | 2.62            |
| 20.0                             | 68.0                             | 25.0025192              | 2.37            |
| 20.5                             | 68.9                             | 25.0025119              | 2.08            |
| 21.0                             | 69.8                             | 25.0025036              | 1.74            |
| 21.5                             | 70.7                             | 25.0024942              | 1.37            |
| 22.0                             | 71.6                             | 25.0024838              | 0.95            |
| 22.5                             | 72.5                             | 25.0024724              | 0.50            |
| 23.0                             | 73.4                             | 25.0024600              | 0.00            |
| 23.5                             | 74.3                             | 25.0024465              | -0.54           |
| 24.0                             | 75.2                             | 25.0024320              | -1.12           |
| 24.5                             | 76.1                             | 25.0024165              | -1.74           |
| 25.0                             | 77.0                             | 25.0024000              | -2.40           |
| 25.5                             | 77.9                             | 25.0023824              | -3.10           |
| 26.0                             | 78.8                             | 25.0023638              | -3.85           |
| 26.5                             | 79.7                             | 25.0023441              | -4.63           |
| 27.0                             | 80.6                             | 25.0023234              | -5.46           |
| 27.5                             | 81.5                             | 25.0023017              | -6.33           |
| 28.0                             | 82.4                             | 25.0022790              | -7.24           |

Resistance vs Temperature

